

Grape juice, but not orange juice or grapefruit juice, inhibits human platelet aggregation.

Coronary artery disease is responsible for much mortality and morbidity around the world. Platelets are involved in atherosclerotic disease development and the reduction of platelet activity by medications reduces the incidence and severity of disease. Red wine and grapes contain polyphenolic compounds, including flavonoids, which can reduce platelet aggregation and have been associated with lower rates of cardiovascular disease. Citrus fruits contain different classes of polyphenolics that may not share the same properties. This study evaluated whether commercial grape, orange and grapefruit juices, taken daily, reduce ex vivo platelet activity. In a randomized cross-over design, ten healthy human subjects (ages 26-58 y, five of each gender) drank 5-7.5 mL/(kg. d) of purple grape juice, orange juice or grapefruit juice for 7-10 d each. Platelet aggregation (whole blood impedance aggregometry, Chronolog Model #590) at baseline was compared to results after consumption of each juice. Drinking purple grape juice for one week reduced the whole blood platelet aggregation response to 1 mg/L of collagen by 77% (from 17.9 +/- 2.3 to 4.0 +/- 6.8 ohms, $P = 0.0002$). Orange juice and grapefruit juice had no effect on platelet aggregation. The purple grape juice had approximately three times the total polyphenolic concentration of the citrus juices and was a potent platelet inhibitor in healthy subjects while the citrus juices showed no effect. The platelet inhibitory effect of the flavonoids in grape juice may decrease the risk of coronary thrombosis and myocardial infarction.